



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion

Executive Summary

- Data collected via SpaceX REST API and Wikipedia web scraping. Data wrangled, cleaned and labelled. EDA performed using SQL and visualizations. Interactive analytics built with Folium maps and Plotly Dash dashboard.
- Four ML models evaluated (Logistic Regression, SVM, Decision Tree, KNN). All achieved 83.33% test accuracy. Logistic Regression identified as best model with cross-validation F1 of 0.8812. KSC LC-39A had the highest launch success rate at 76.9%.



Introduction

- **Background**
- SpaceX (Space Exploration Technologies Corp.) is an American aerospace manufacturer founded by Elon Musk in 2002, with the primary goal of reducing space transportation costs and enabling Mars colonization. A key innovation is the Falcon 9 rocket, designed for first-stage reusability - allowing SpaceX to advertise launches at \$62 million compared to competitors charging upward of \$165 million.
- **Explore**
- This project evaluates whether the Falcon 9 first stage will successfully land after launch. By analyzing historical launch data collected via the SpaceX REST API and Wikipedia web scraping, we apply exploratory data analysis, interactive visual analytics, and machine learning classification models to predict landing outcomes. Successfully predicting a landing allows accurate cost estimation - valuable intelligence for any company bidding against SpaceX for a commercial rocket launch contract.

Problems We Aim to Answer

- Which launch site has the highest success rate?
- Which payload range and booster version performs best?
- Can we build a model to predict first-stage landing success with >80% accuracy?

Section 1

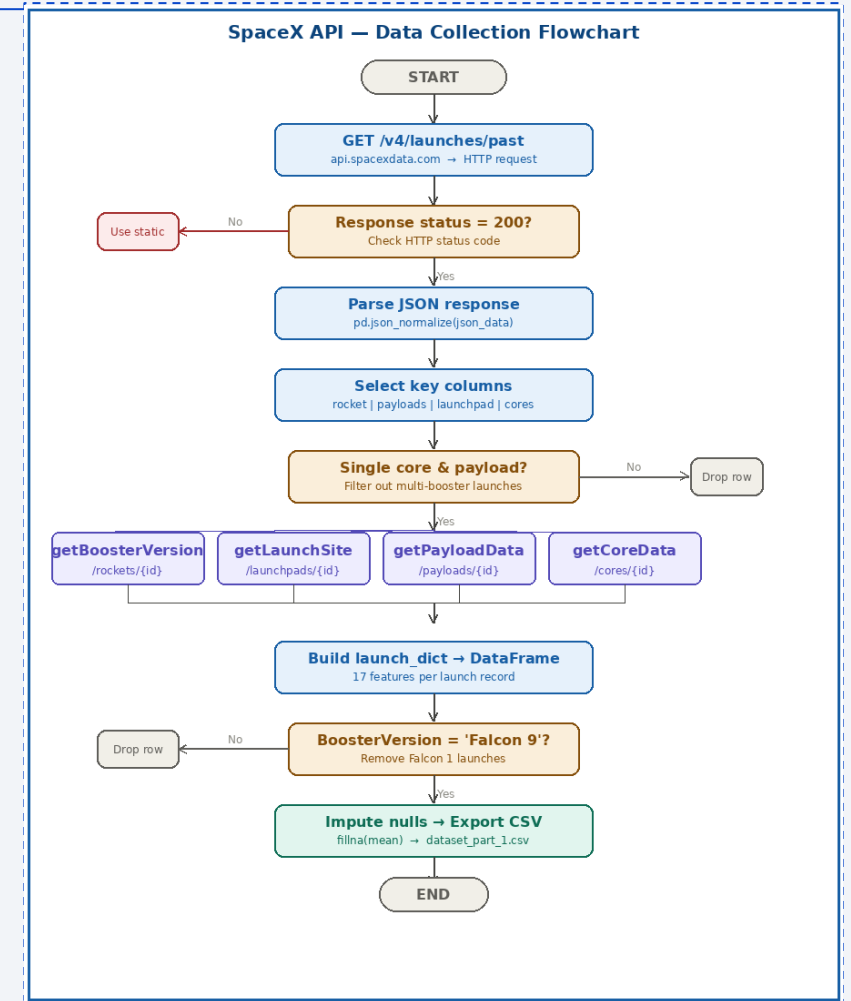
Methodology

Data Collection

- Two datasets were collected: launch records from the SpaceX REST API (v4) and Falcon 9 launch history scraped from Wikipedia. Both datasets were merged and cleaned to form the final training dataset of ~90 Falcon 9 launches.
- API: GET requests to api.spacexdata.com/v4/launches/past. Web Scraping: BeautifulSoup parsed Wikipedia Falcon 9 launch records table. Features extracted: FlightNumber, PayloadMass, Orbit, LaunchSite, Outcome, GridFins, Reused, Legs, Block, Serial, Longitude, Latitude.

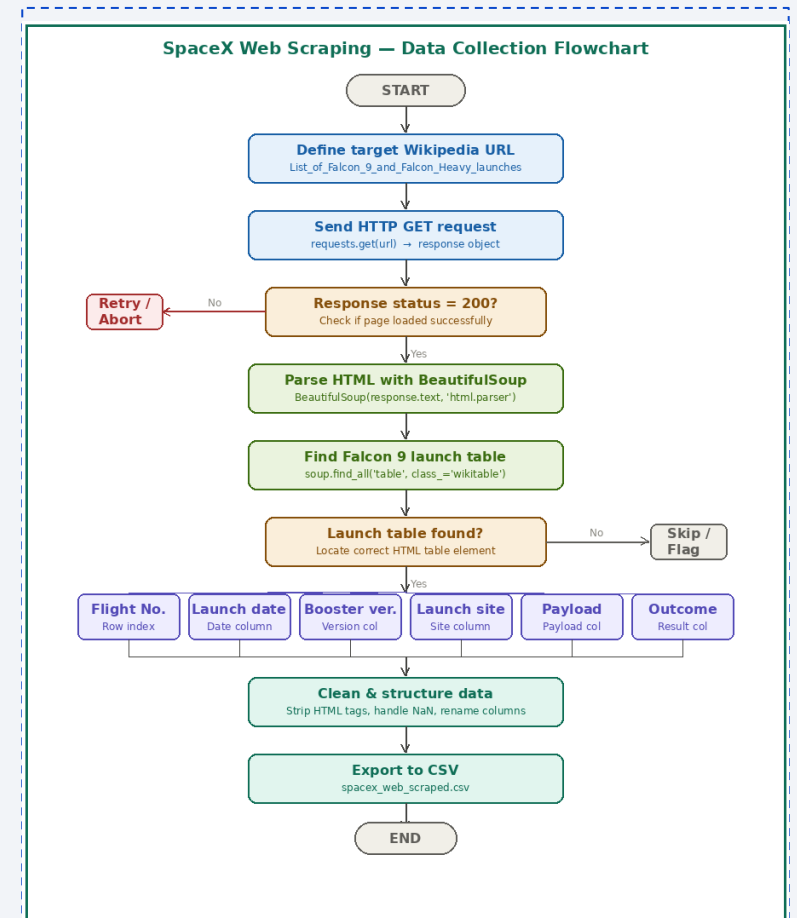
Data Collection – SpaceX API

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- API: GET requests to `api.spacexdata.com/v4/launches/past`. Web Scraping: BeautifulSoup parsed Wikipedia Falcon 9 launch records table. Features extracted: FlightNumber, PayloadMass, Orbit, LaunchSite, Outcome, GridFins, Reused, Legs, Block, Serial, Longitude, Latitude.
- https://github.com/BoldwinMax/SpaceX-Falcon9-Landing-Prediction/blob/main/O1_SpaceX_Data_Collection_API.ipynb



Data Collection - Scraping

- Wikipedia Falcon 9 launch records table scraped using BeautifulSoup and Requests. Parsed HTML table columns: Flight No., Date, Booster version, Launch site, Payload, Orbit, Customer, Launch outcome, Landing outcome. Data extracted into a Pandas DataFrame and exported as CSV.
- https://github.com/BoldwinMax/SpaceX-Falcon9-Landing-Prediction/blob/main/02_SpaceX_Data_Web scraping.ipynb

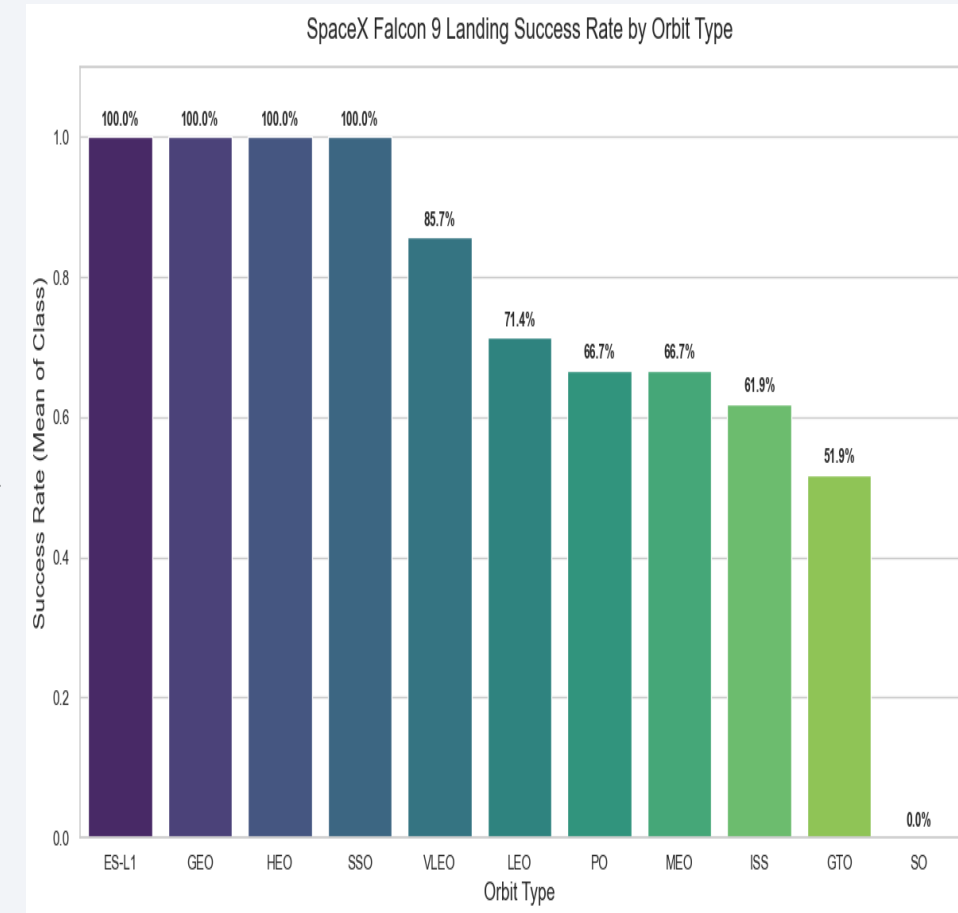


Data Wrangling

- Missing values handled: PayloadMass imputed with column mean (5 nulls). LandingPad retained None values (26 nulls - no landing pad used). Binary class label created: 1 = successful landing, 0 = failed. Falcon 1 launches filtered out, keeping only Falcon 9.
- Process: Load JSON - Filter Falcon 9 - Extract features via API - Handle nulls - Create class label - Export CSV
- GitHub: https://github.com/BoldwinMax/SpaceX-Falcon9-Landing-Prediction/blob/main/notebooks/03_data_wrangling.ipynb

EDA with Data Visualization

- Charts plotted: Scatter plot of Flight Number vs Launch Site (to see launch distribution). Payload Mass vs Launch Site (to find payload patterns). Success Rate vs Orbit Type (bar chart). Flight Number vs Orbit Type. Payload vs Orbit. Yearly success trend line chart. Charts revealed KSC LC-39A as most active and successful site, and FT booster as most reliable.
- [GitHub: https://github.com/BoldwinMax/SpaceX-Falcon9-Landing-Prediction/blob/main/notebooks/05_eda_visualization.ipynb](https://github.com/BoldwinMax/SpaceX-Falcon9-Landing-Prediction/blob/main/notebooks/05_eda_visualization.ipynb)



EDA with SQL

- SQL queries performed: All unique launch site names. Sites beginning with CCA. Total payload mass from NASA boosters. Average payload mass for F9 v1.1. First successful ground landing date. Drone ship landings with payload 4000-6000 kg. Total successful vs failed missions. Booster with max payload. 2015 launch records with failed drone ship landings. Landing outcome rankings 2010-2017.
- [GitHub: https://github.com/BoldwinMax/SpaceX-Falcon9-Landing-Prediction/blob/main/notebooks/04_eda_sql.ipynb](https://github.com/BoldwinMax/SpaceX-Falcon9-Landing-Prediction/blob/main/notebooks/04_eda_sql.ipynb)

Average_Payload_Mass_F9_v1.1

2534.6666666666665

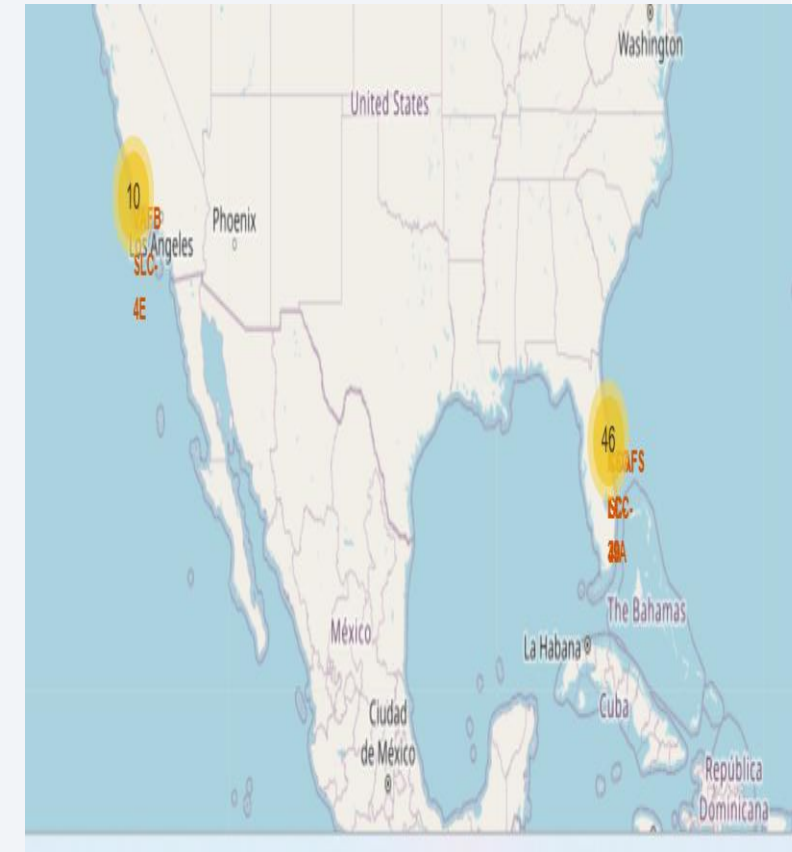
First_Successful_Ground_Pad_Landing

2015-12-22

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Build an Interactive Map with Folium

- Map objects created: Circle markers for all 4 launch sites with site name popups . Color-coded MarkerCluster: green = success, red = failure for each launch record . MousePosition plugin for live coordinate tracking. PolyLines from launch sites to closest coastline, city, railway, and highway with distance labels .
- Findings: All launch sites are near the coast and equator. KSC LC-39A and CCAFS SLC-40 are within 1 km of the coastline. Sites are near highways and railways but far from cities - a safety requirement.
- GitHub: https://github.com/BoldwinMax/SpaceX-Falcon9-Landing-Prediction/blob/main/notebooks/O6_interactive_map_folium.ipynb

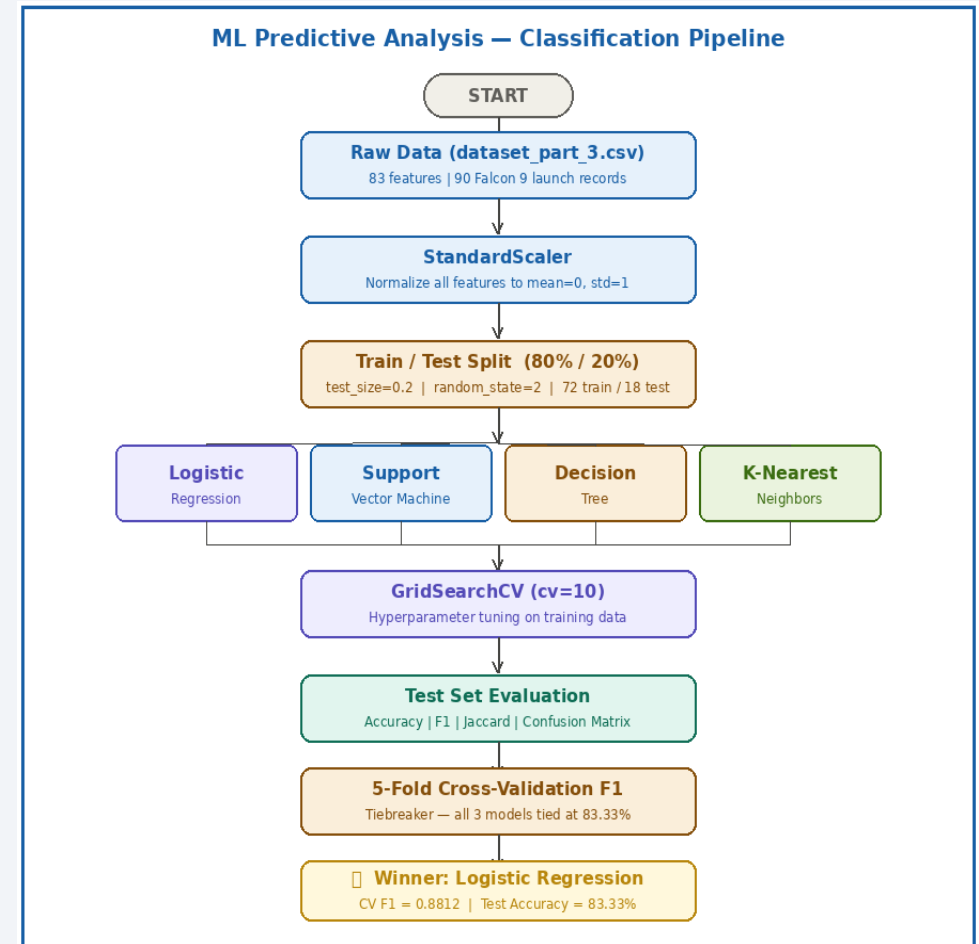


Build a Dashboard with Plotly Dash

- Dashboard components: Dropdown to filter by launch site (All Sites or individual site). Pie chart showing total successful launches by site (All Sites) or success vs failure split per site. Range slider (0-10,000 kg) to filter payload mass. Scatter plot showing Payload Mass vs Launch Outcome color-coded by Booster Version Category.
- Insights enabled: Identify which site has highest success rate. See how payload mass correlates with outcome. Compare booster version performance across payload ranges.
- https://github.com/BoldwinMax/SpaceX-Falcon9-Landing-Prediction/blob/main/07_SpaceX_Visual_Analytics_Plotly_app.py

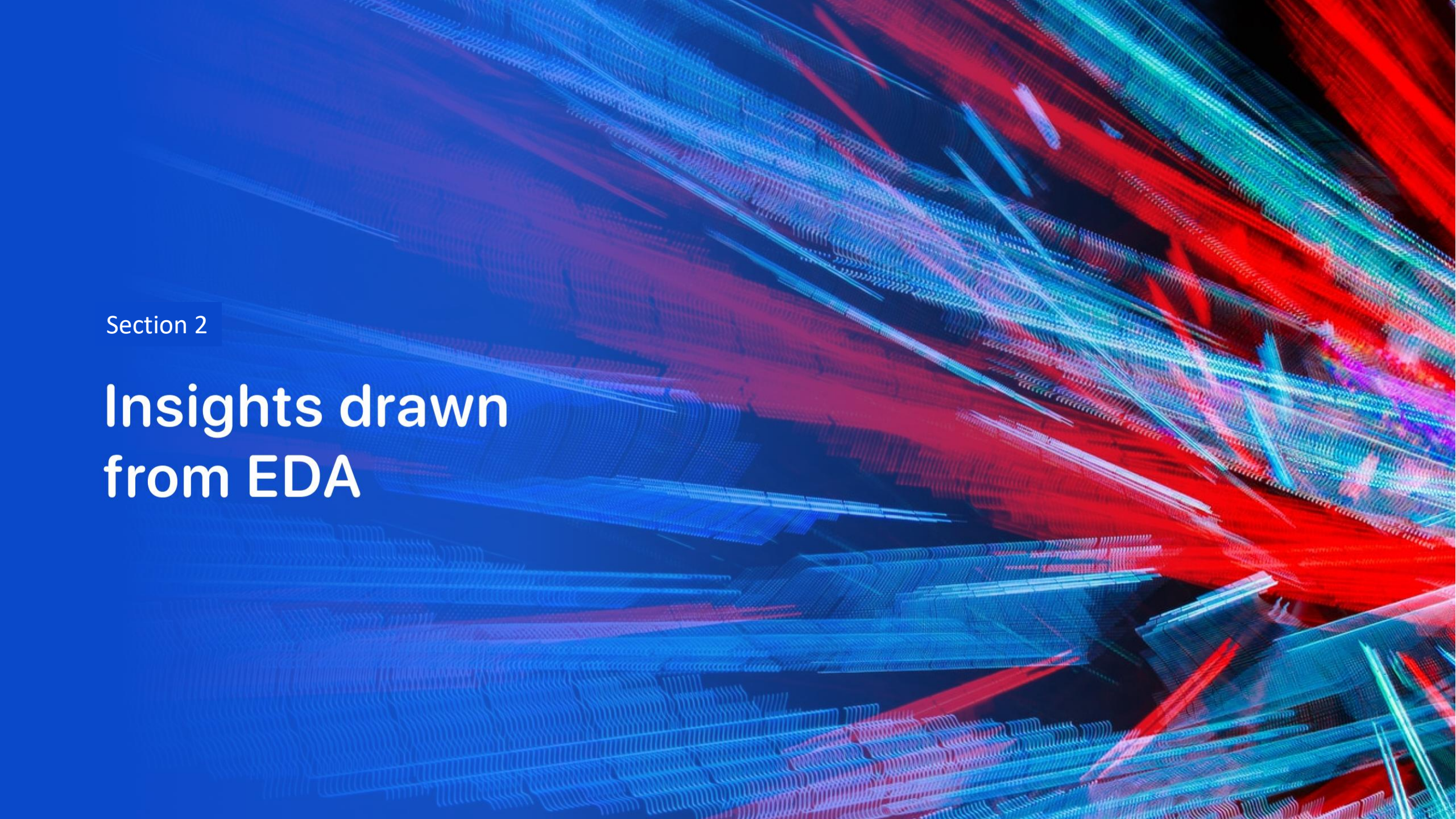
Predictive Analysis (Classification)

- Pipeline: Standardized features with StandardScaler. Split 80/20 train/test (random_state=2). Trained 4 models: Logistic Regression, SVM, Decision Tree, KNN. Used GridSearchCV (cv=10) for hyperparameter tuning. Evaluated with accuracy, F1, Jaccard scores, confusion matrices and 5-fold cross-validation F1 as tiebreaker.
- GitHub:
https://github.com/BoldwinMax/SpaceX-Falcon9-Landing-Prediction/blob/main/notebooks/08_predictive_analysis_classification.ipynb



Results

- EDA revealed KSC LC-39A as top launch site (41.7% of successes). Success rates improved dramatically after 2013. FT booster version most reliable across all payload ranges.
- Folium maps confirmed coastal proximity of all launch sites. Plotly Dash dashboard showed payload 2,000-4,000 kg has highest success rate.
- Three of the 4 models achieved 83.33% test accuracy. Logistic Regression won with CV F1 = 0.8812. Decision Tree was weakest (CV F1 = 0.8058, highest variance +/-0.0916).

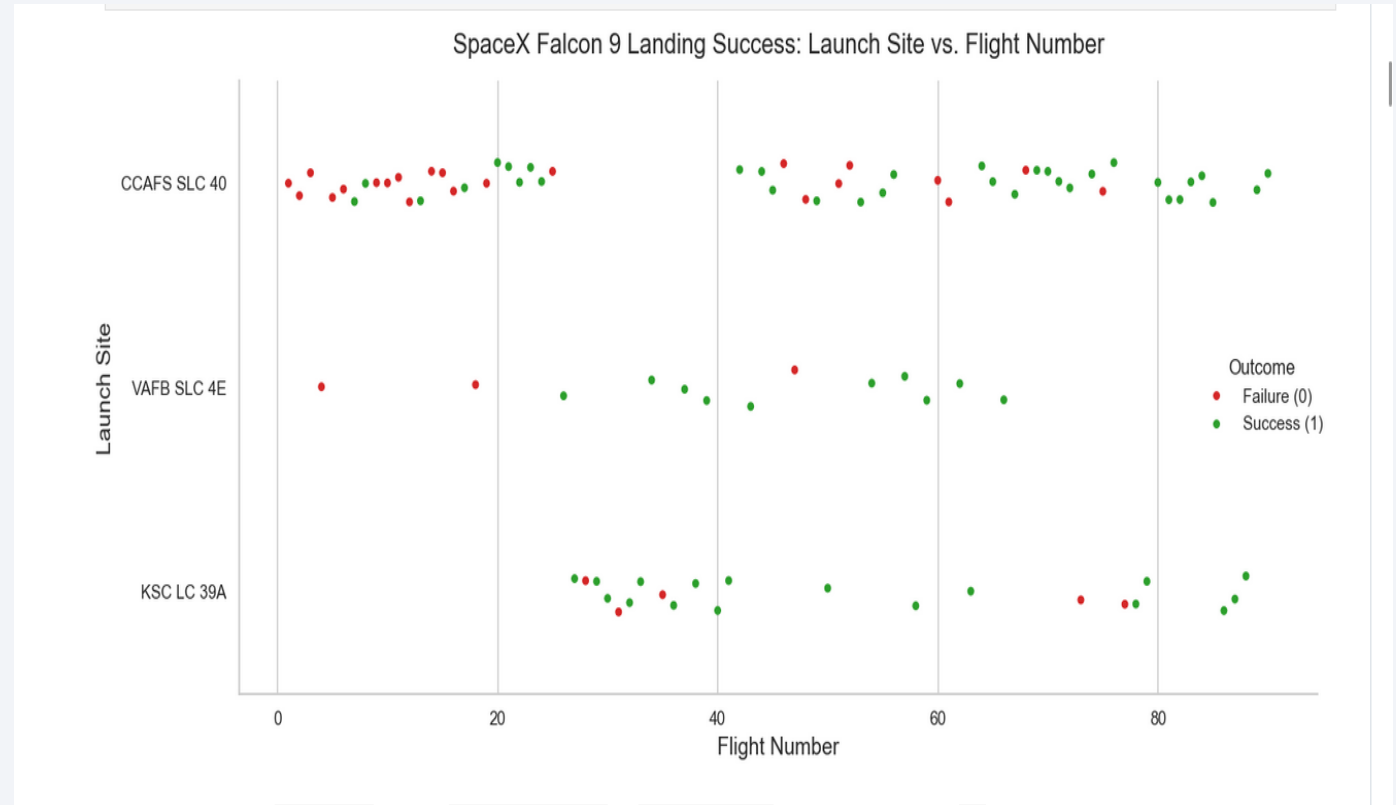
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

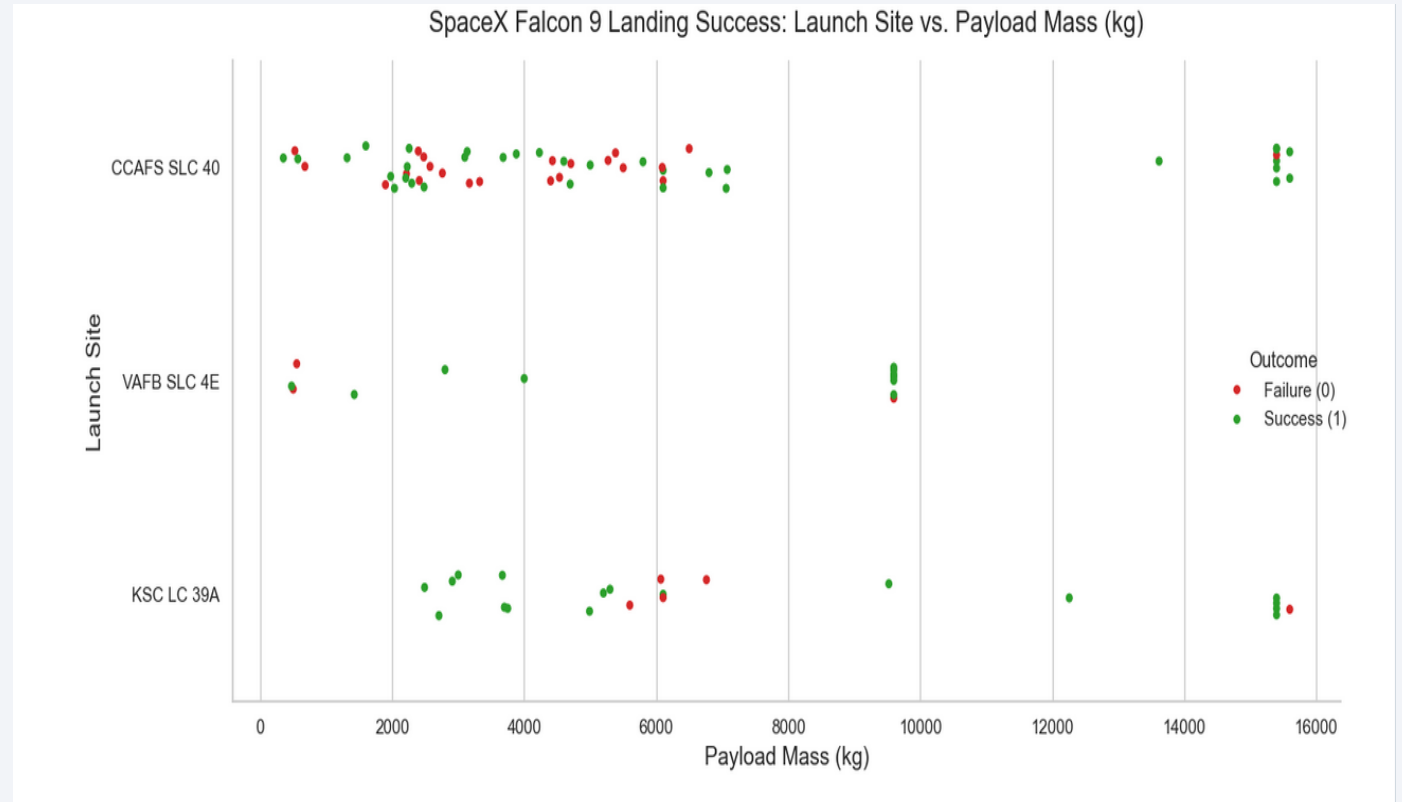
Flight Number vs. Launch Site

- CCAFS SLC-40 dominated early launches (flights 1-50). KSC LC-39A became dominant from flight 60+. Launch success rates improved steadily with each flight, confirming SpaceX improved its landing technology over time.
- Later flight numbers (60+) show higher success rates across all sites, confirming that SpaceX improved its landing technology over time.



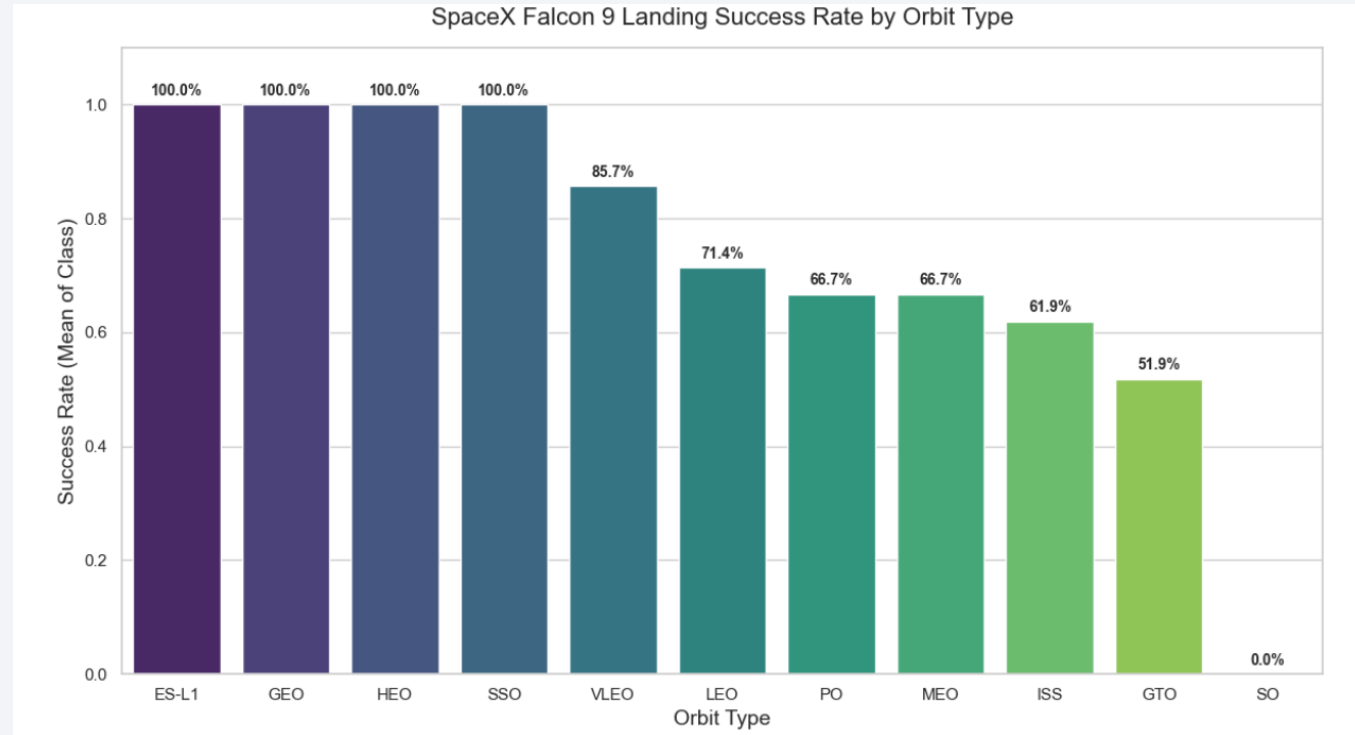
Payload vs. Launch Site

- Payload Mass vs Launch Site: KSC LC-39A handles widest range (2,000-16,000 kg). VAFB SLC-4E limited to lighter payloads. CCAFS SLC-40 handles medium payloads. KSC LC-39A shows success across all sizes.
- KSC LC-39A shows success across all payload sizes. Heavier payloads (>10,000 kg) are exclusively handled by KSC LC-39A.



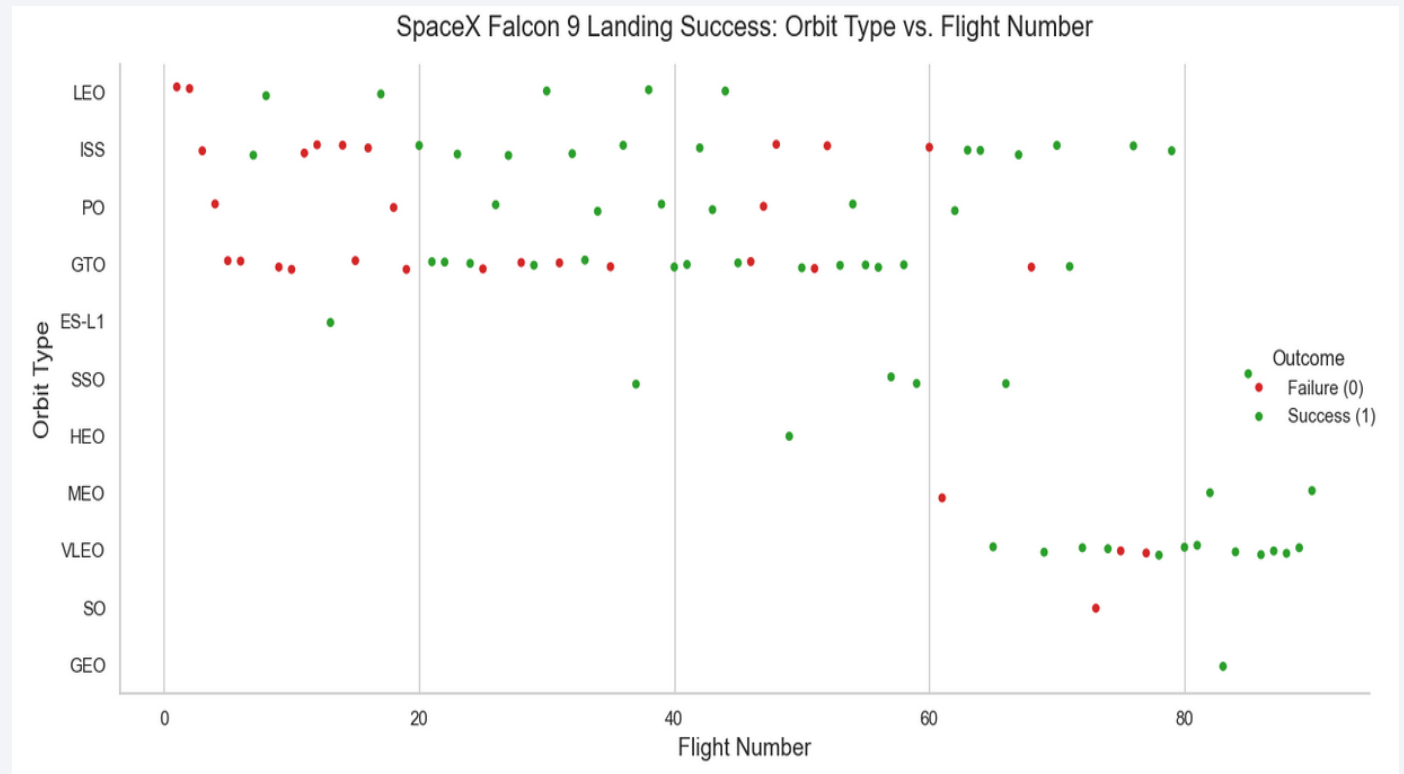
Success Rate vs. Orbit Type

- ES-L1, GEO, HEO, SSO orbits achieved 100% success rate. LEO and GTO had lower success rates. Higher orbits used improved FT boosters. MEO and POL showed high success rates.
- Higher orbit types (ES-L1, GEO, HEO, SSO) showed 100% success - these later missions used improved FT booster technology.



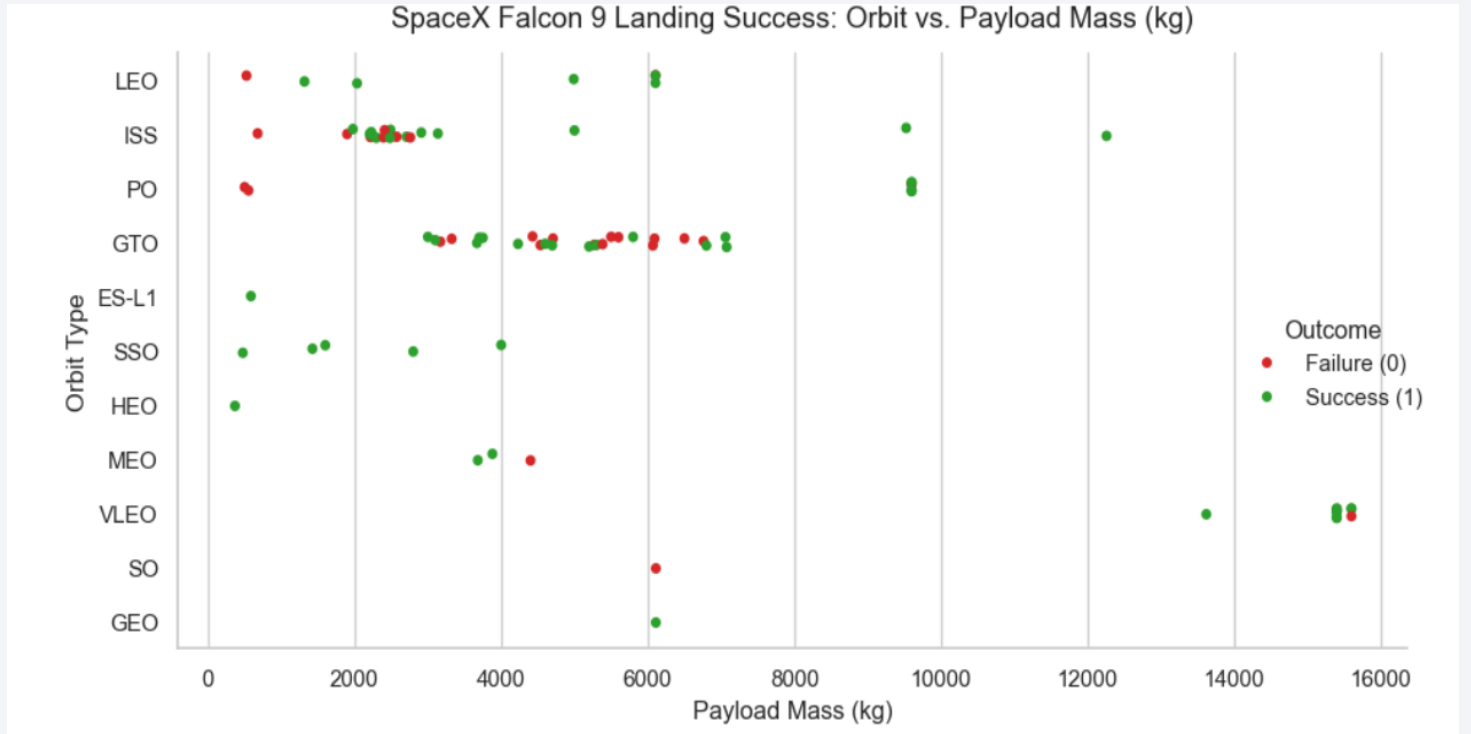
Flight Number vs. Orbit Type

- Early flights (1-20) concentrated on LEO/ISS missions. Later flights expanded to GTO, VLEO, SSO. VLEO (Starlink) missions dominated from flight 80+ showing orbital diversity growth.
- Orbit diversity increased with flight number, showing SpaceX expanded its commercial capabilities over time.



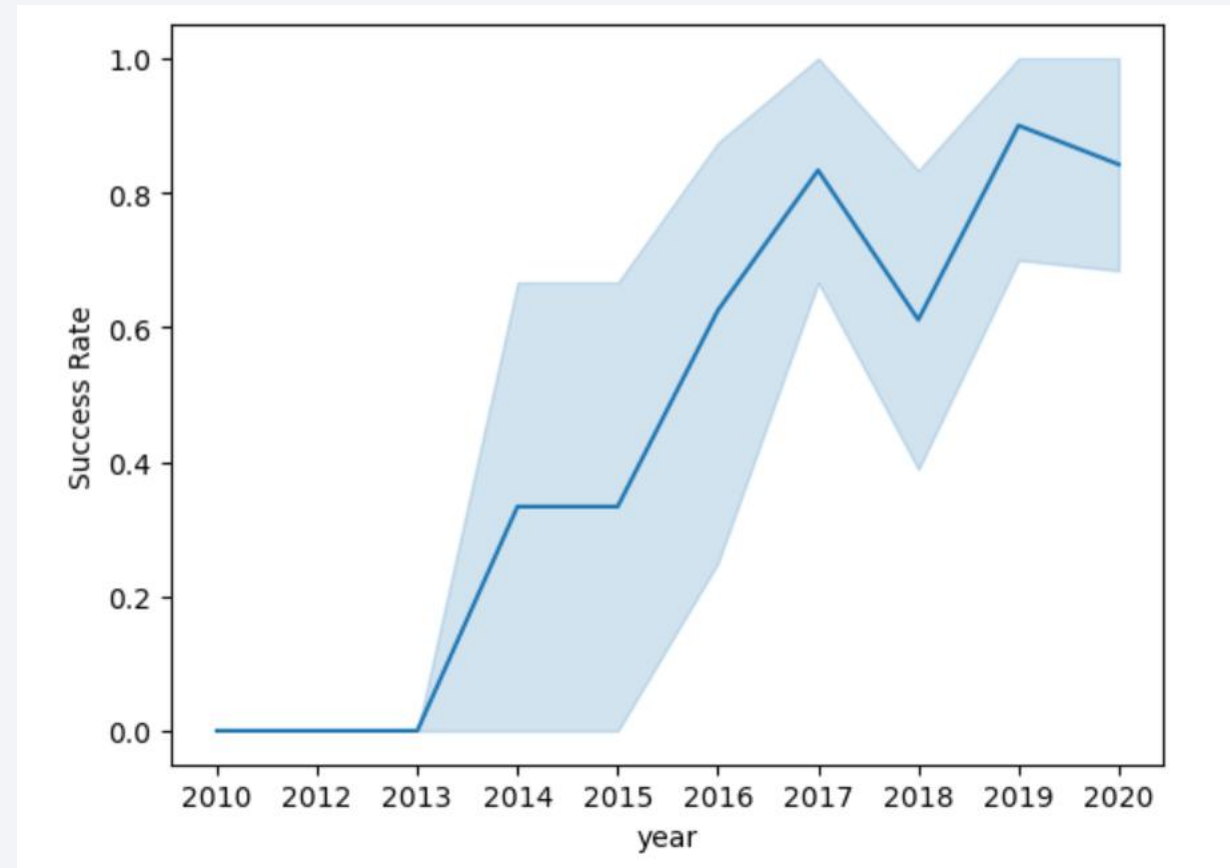
Payload vs. Orbit Type

- VLEO missions (Starlink) carry heaviest payloads (15,000+ kg). GTO carries 3,000-6,000 kg. LEO and ISS vary widely. SSO missions tend to be lighter. Payload strongly correlates with orbit type.
- Payload mass strongly correlates with orbit type. VLEO/Starlink missions consistently carry maximum payloads and show increasing success rates.



Launch Success Yearly Trend

- Line chart of Launch Success Yearly Trend: from 0% in 2010 to 80%+ by 2017 and near 100% by 2020. Each year shows consistent improvement as SpaceX refined booster recovery technology.
- SpaceX achieved operational maturity by 2017-2018. Landing prediction is most reliable for recent launches.



All Launch Site Names

- CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
- SpaceX operates 4 unique launch sites across 2 US coastal locations: Cape Canaveral Air Force Station (FL) and Vandenberg Air Force Base (CA)

```
Out[11]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

Launch Site Names Begin with 'CCA'

- 5 records from CCAFS LC-40 and CCAFS SLC-40. Both sites are at Cape Canaveral Air Force Station, Florida - a key SpaceX launch location for eastward equatorial orbits.

Out[12]:									
Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	

Total Payload Mass

- 45,596 kg total payload delivered to orbit by SpaceX boosters for NASA. This demonstrates SpaceX's critical role as a NASA Commercial Resupply Services provider.

```
sqlite:///my_data1.db
Done.
Out[13]: Total_Payload_Mass_NASA_CRS
         45596
```

Average Payload Mass by F9 v1.1

- 2,534.7 kg average payload. The F9 v1.1 was SpaceX's second Falcon 9 version, used 2013-2015 before the more powerful Falcon 9 Full Thrust (FT) replaced it.

```
* sqlite:///my_data1.db  
Done.
```

```
Out[14]: Average_Payload_Mass_F9_v1.1  
2534.6666666666665
```


First Successful Ground Landing Date

- 2015-12-22. This historic date marks when SpaceX first successfully landed a Falcon 9 first stage at Landing Zone 1, Cape Canaveral - a milestone in reusable rocket history.

```
DONE.  
  
Out[15]: First_Successful_Ground_Pad_Landing  
  
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2. All successful drone ship landings in this range used the reliable FT booster version.

Out[16]: **Booster_Version**

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- The vast majority of Falcon 9 missions (98 out of 101) resulted in a successful outcome
- Only 1 in-flight failure was recorded - the CRS-7 mission in June 2015
- Overall mission success rate is approximately 99%, demonstrating SpaceX's reliability in delivering payloads to orbit.

Done.

Out[17]:

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- All maximum payload carriers are Block 5 (F9 B5) boosters, SpaceX's most advanced and final Falcon 9 version
- The 15,600 kg payload corresponds to Starlink satellite batch deployments - SpaceX's own internet constellation missions
- Block 5 was specifically designed for heavy payloads and rapid reusability, capable of flying the same booster 10+ times
- This confirms that the B5 booster is SpaceX's most capable and reliable version for heavy-lift missions

Out[18]:

Booster_Version	PAYLOAD_MASS_KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

- Both 2015 drone ship failures used the F9 v1.1 booster - SpaceX's second generation, not yet refined for reliable recovery
- Both failures launched from CCAFS LC-40, Cape Canaveral
- 2015 was a critical learning year - SpaceX achieved its first successful ground landing in December 2015, just months after these drone ship failures
- These failures directly contributed to the engineering improvements that made the FT booster so successful from 2016 onwards

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Out[19]:
```

Month_Name	Landing_Outcome	Booster_Version	Launch_Site
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January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
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April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40
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Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- "No attempt" dominates (10) , early SpaceX missions made no landing attempt at all, focusing entirely on payload delivery
- Drone ship success and failure are tied at 5 each, showing SpaceX was still mastering sea landing in this period
- Only 3 ground pad successes recorded - these began in December 2015
- This early era (2010-2017) represents SpaceX's learning phase, which explains the lower ML model accuracy for older launches

Done.

Landing_Outcome	count_outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

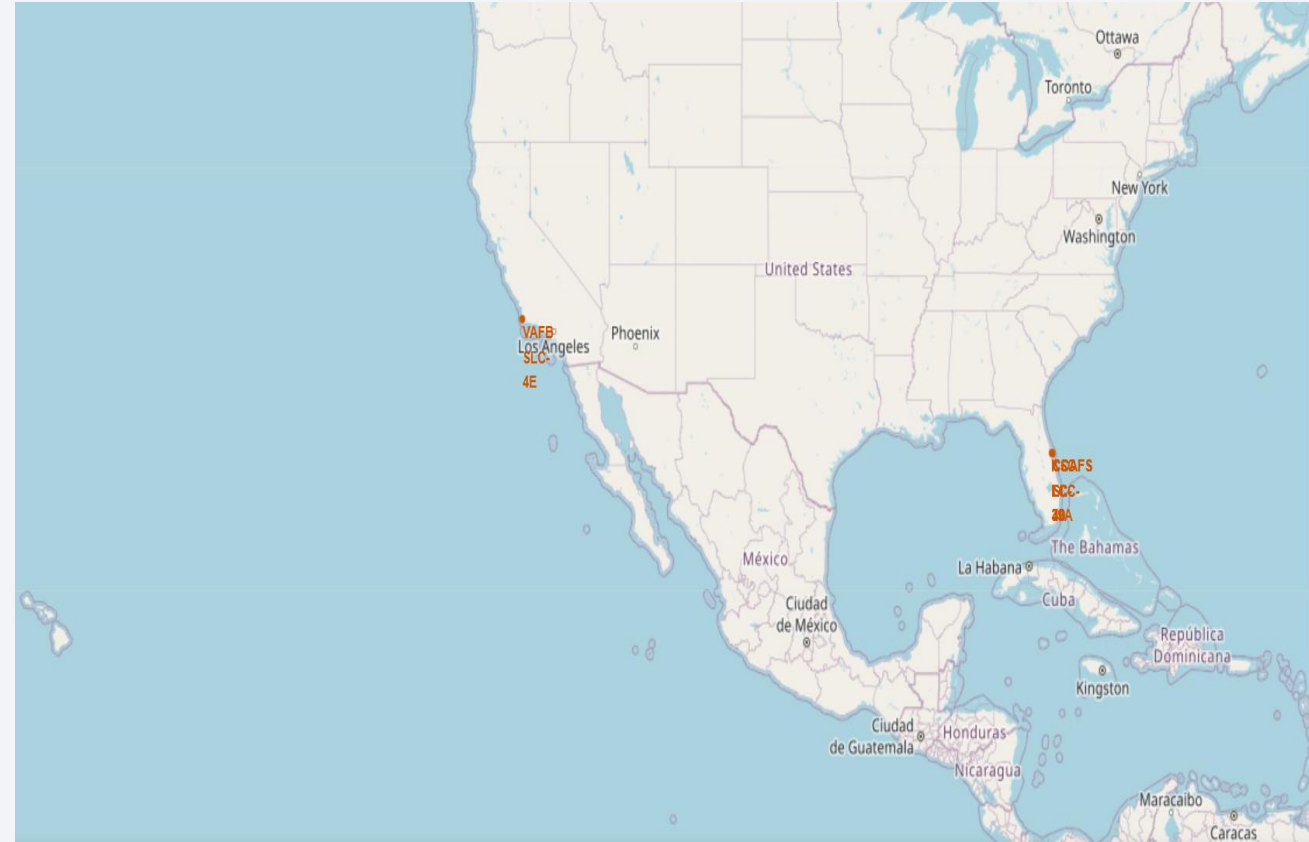
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left shows a clear blue sky.

Section 3

Launch Sites Proximities Analysis

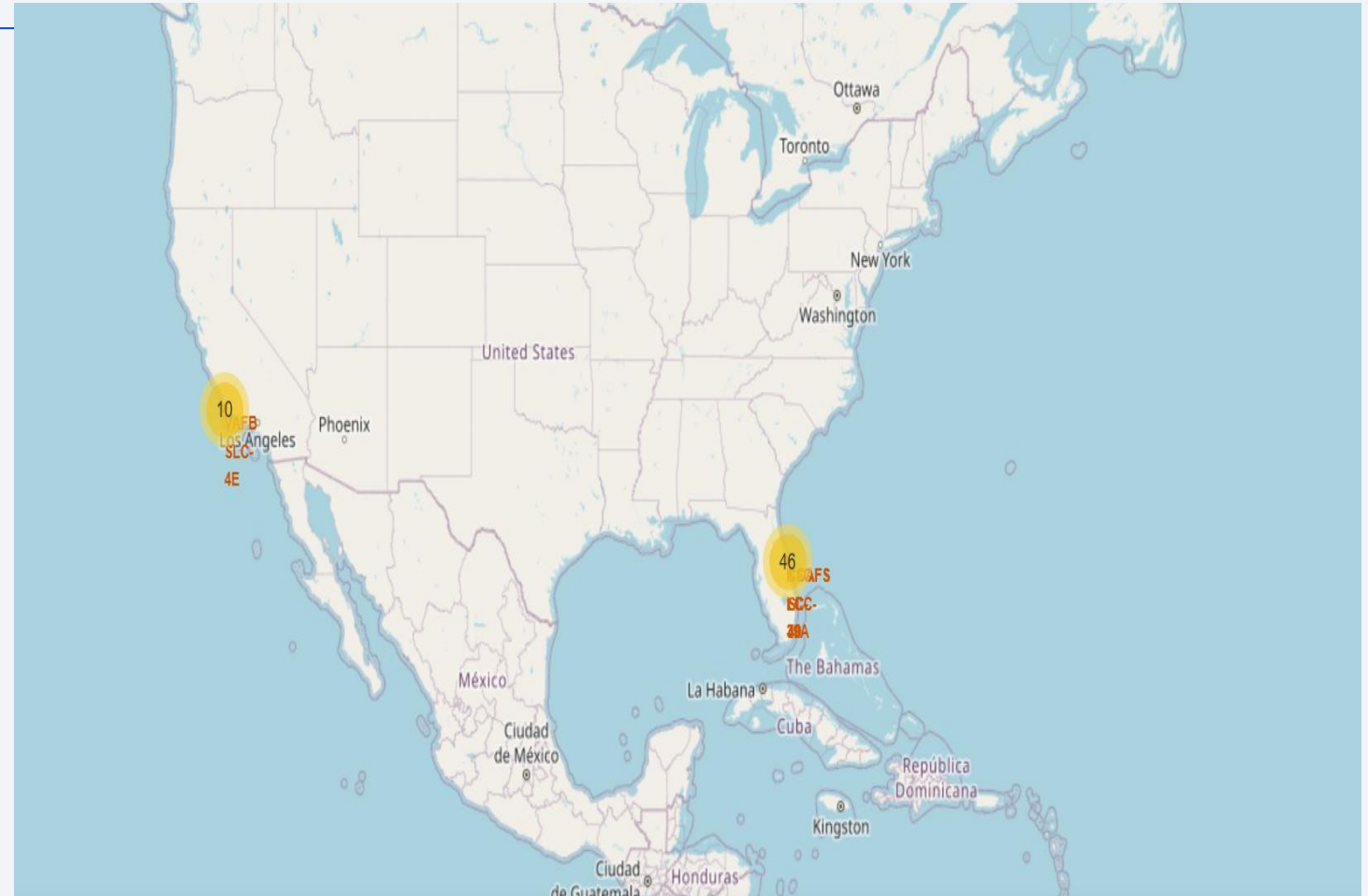
All SpaceX Launch Sites - Global Location Map

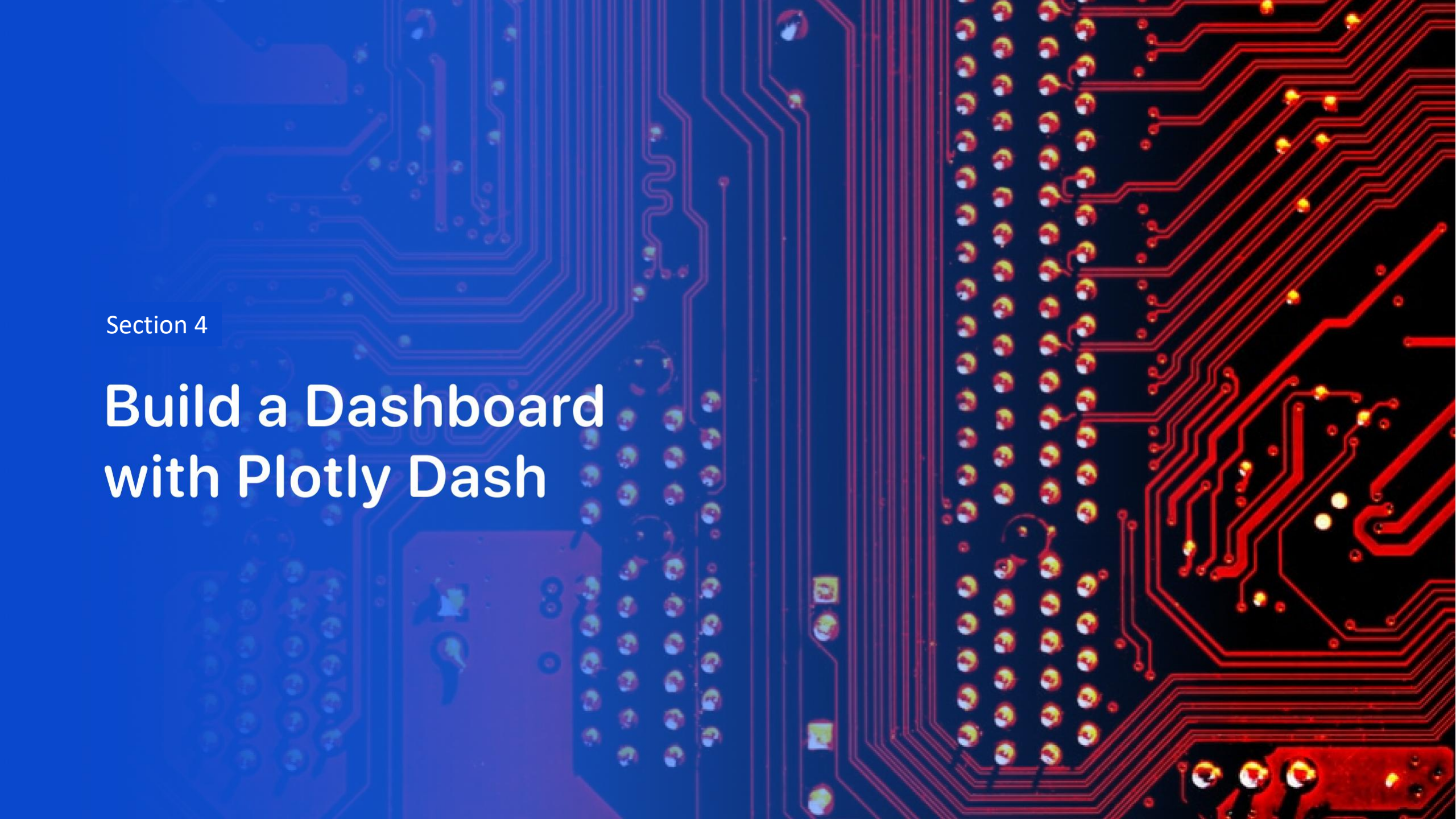
- 4 launch sites marked: VAFB SLC-4E (California) and CCAFS LC-40, CCAFS SLC-40, KSC LC-39A (Florida)
- All sites are on coastlines - rockets launch over ocean for safety
- Florida sites handle equatorial orbits, California handles polar orbits



Launch Outcomes - Color-Coded Marker Clusters

- Yellow clusters show grouped launch records - Florida (46 launches) and California (10 launches)
- Zoom in reveals green markers (success) vs red markers (failure) per launch
- Florida cluster is larger confirming it as SpaceX's primary launch region



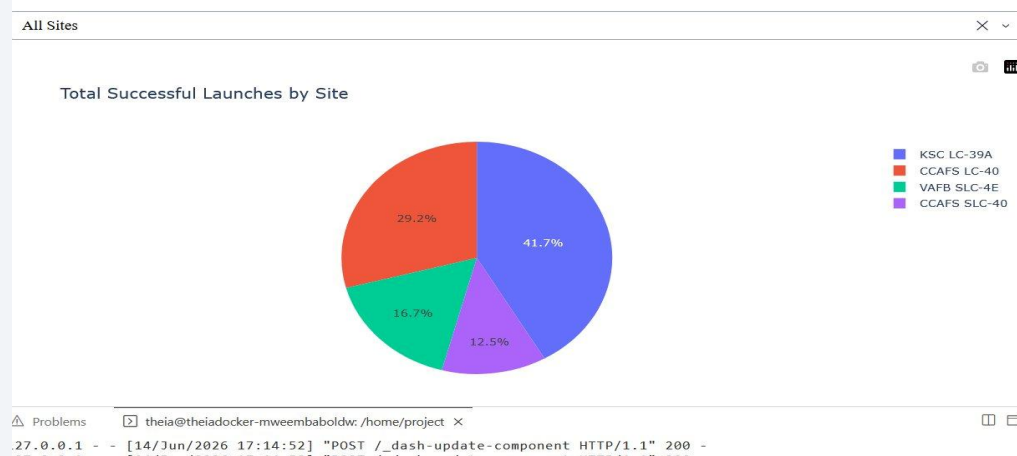


Section 4

Build a Dashboard with Plotly Dash

All Sites Pie Chart - KSC LC-39A Leads at 41.7%

Total Successful Launches by Site -All Sites

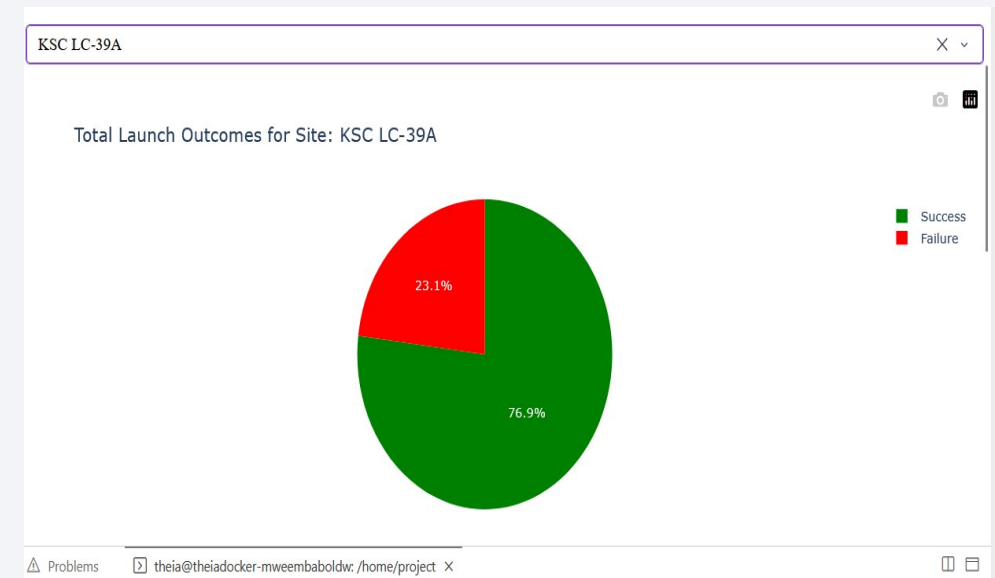


- Pie chart showing: KSC LC-39A = 41.7% (largest share), CCAFS LC-40 = 29.2%, VAFB SLC-4E = 16.7%, CCAFS SLC-40 = 12.5%. Dropdown set to All Sites. Interactive pie chart from Plotly Dash dashboard.
- KSC LC-39A accounts for 41.7% of all successful launches -confirming it as SpaceX's most productive and reliable launch site.

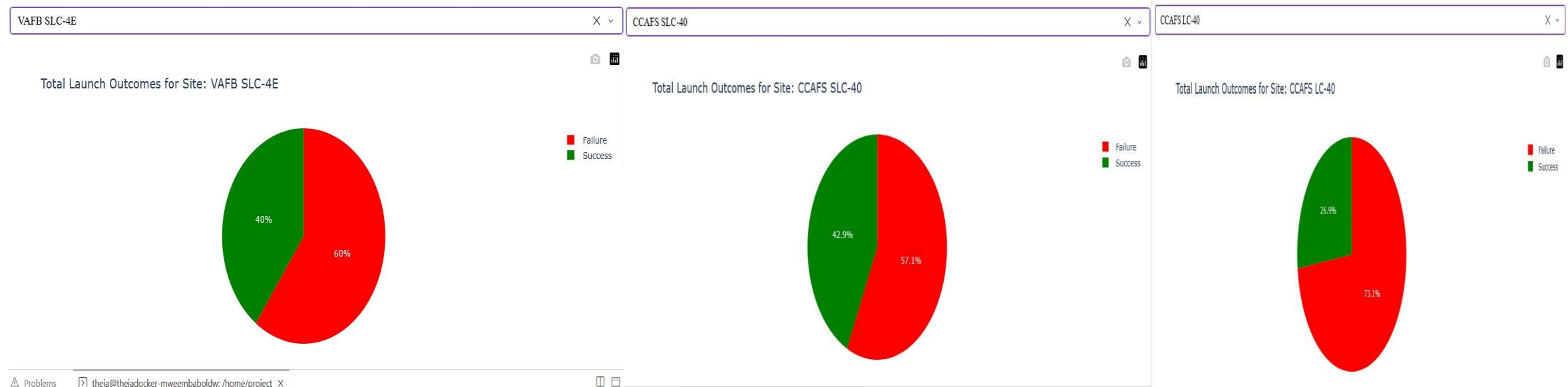
KSC LC-39A Success Rate: 76.9%

- KSC LC-39A site-specific pie chart: Success = 76.9% (green), Failure = 23.1% (red). This is the highest success rate of all 4 sites. Comparison: CCAFS LC-40 = 26.9%, CCAFS SLC-40 = 42.9%, VAFB SLC-4E = 40.0%.
- KSC LC-39A has the highest success rate at 76.9%. It handles the widest payload range and largest missions including Starlink deployments.

KSC LC-39A Highest Launch Success Rate



Individual Site Success Rates



Key findings from site-by-site analysis:

VAFB SLC-4E - 40% success rate (60% failure). California-based site handles polar orbit missions exclusively. Lower success rate reflects earlier-era launches before FT booster improvements.

CCAFS SLC-40 -42.9% success rate (57.1% failure). One of SpaceX's earliest active sites. Improved significantly after 2017 when the FT booster was introduced.

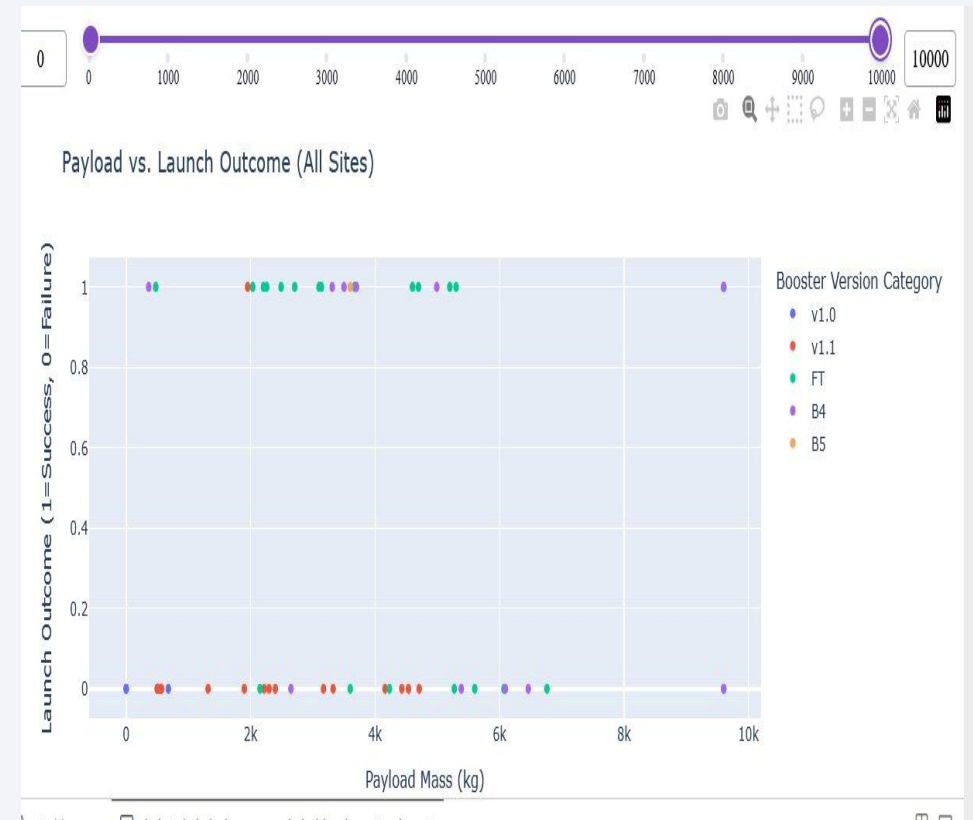
CCAFS LC-40- 26.9% success rate (73.1% failure). Lowest success rate of all sites. Predominantly used for early Falcon 9 missions when landing technology was still being developed.

Overall insight: All three sites show majority failure rates, confirming that KSC LC-39A (76.9% success) stands out as SpaceX's most reliable and mature launch site -used for the newest, most successful missions including Starlink deployments.

Scatter Chart - FT Booster Most Reliable Across Payload Ranges

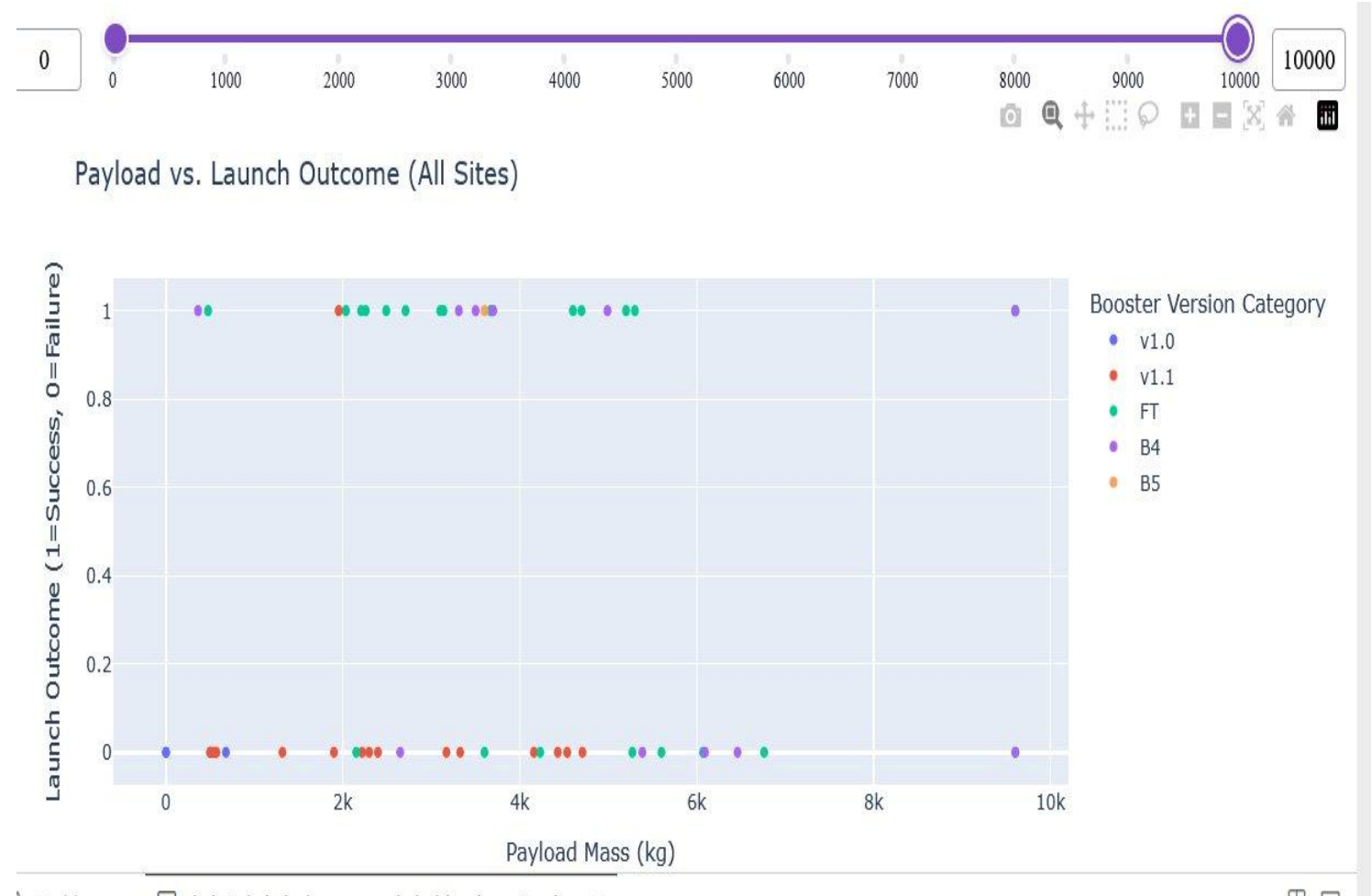
- Scatter plot: X=Payload Mass (kg), Y=Launch Outcome (1=Success, 0=Failure). Points color-coded by Booster Version: v1.0 (blue), v1.1 (red), FT (green), B4 (purple), B5 (orange). Slider set to full range 0-10,000 kg.tes, with different payload selected in the range slider
- FT booster (green) dominates successes across 2,000-6,000 kg range. v1.0 and v1.1 (early versions) mostly failures. B4/B5 show high success rates. Payload range 2,000-4,000 kg has highest success rate. which payload range or booster version have the largest success rate, etc.

Payload vs. Launch Outcome - All Sites, All Boosters



Scatter Chart –Payload vs. Launch Outcome - All Sites, All Booster Versions

- FT booster (green) dominates successes across 2,000–6,000 kg - most reliable version
- Early boosters v1.0 and v1.1 (blue/red) mostly failures - landing tech still developing
- Payload range 2,000-5,000 kg has the highest success density across all booster versions



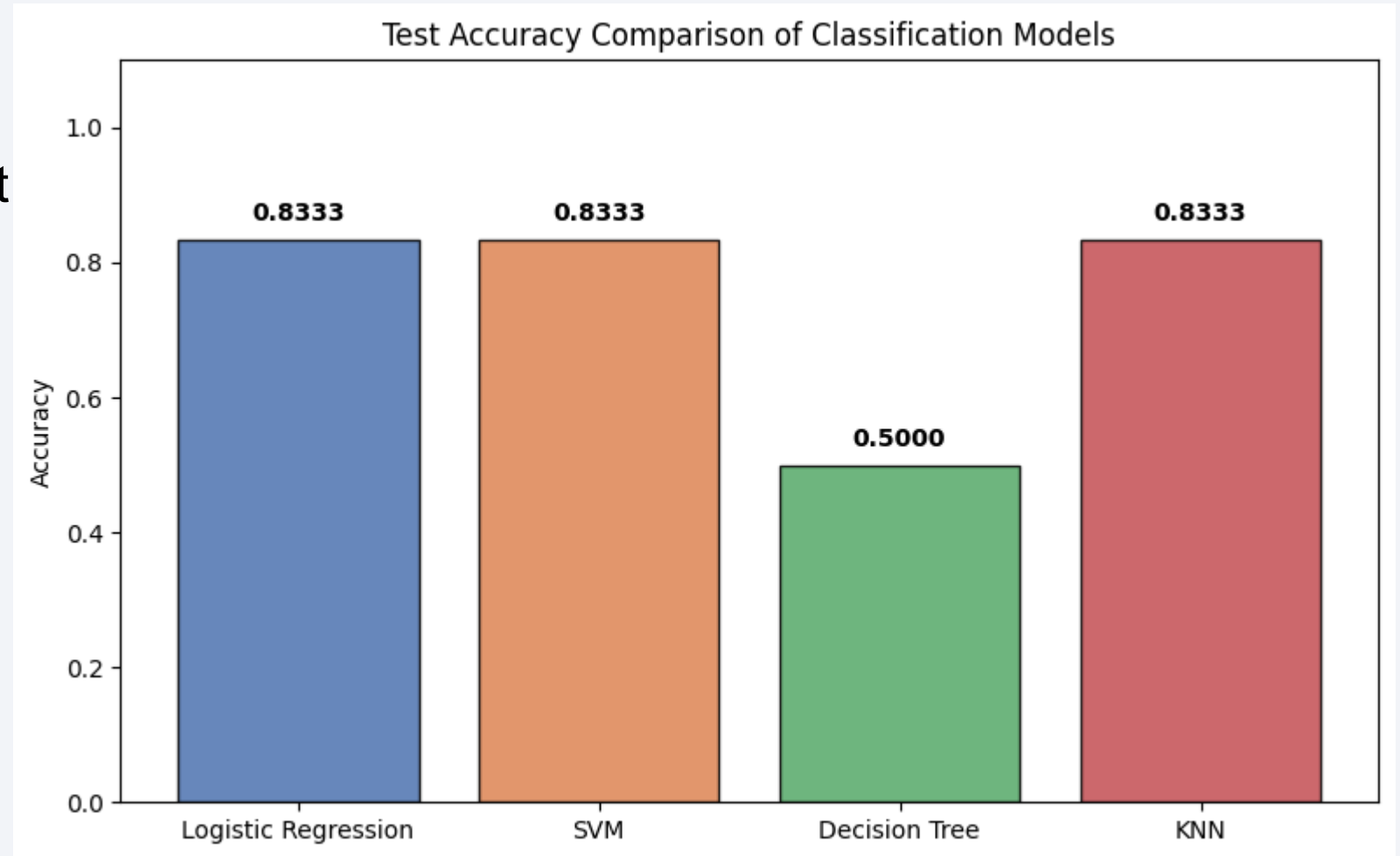


Section 5

Predictive Analysis (Classification)

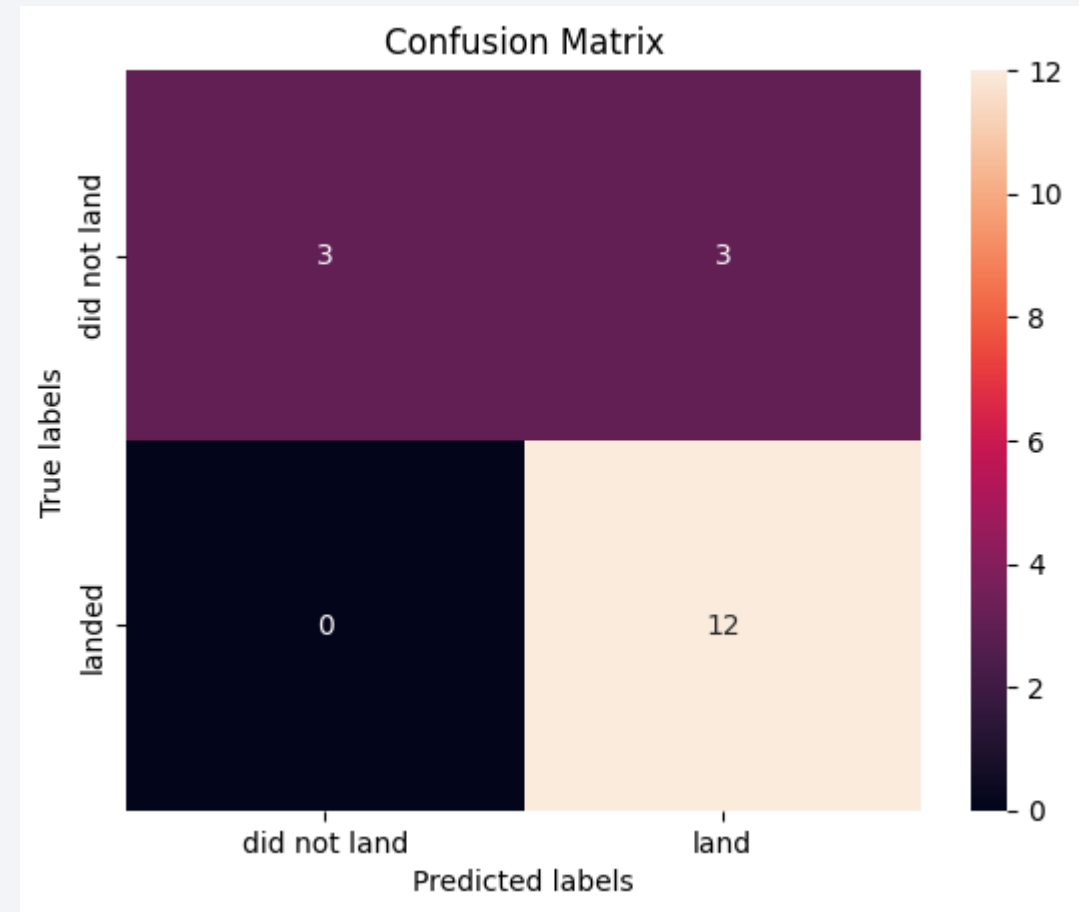
Classification Accuracy

- Logistic Regression, SVM and KNN all tied at **83.33%** test accuracy - Decision Tree scored lowest at **50.00%**
- Tiebreaker using 5-fold cross-validation F1: **Logistic Regression wins (0.8812)**, confirming it as the best overall model



Confusion Matrix

- **True Positives (landed, predicted landed): 12** -model correctly identified all actual landings
- **True Negatives (did not land, predicted did not land): 3** correctly identified failures
- **False Positives (did not land, predicted landed): 3** only errors in the model
- **False Negatives: 0** the model missed zero actual landings (Recall = 100%)



Conclusions

- KSC LC-39A is SpaceX's best launch site with 76.9% success rate and 41.7% of all successful launches
- Launch success rate improved dramatically from 0% (2010) to 90%+ (2020) - SpaceX achieved operational maturity by 2017
- FT booster version performs best; payload range 2,000-4,000 kg has the highest launch success rate
- Logistic Regression is the best ML model (CV F1=0.8812, Test Accuracy=83.33%) - simpler models can outperform complex ones on small datasets
- All launch sites are coastal and near infrastructure but distant from cities ,a deliberate safety and logistics design choice

Thank you!

